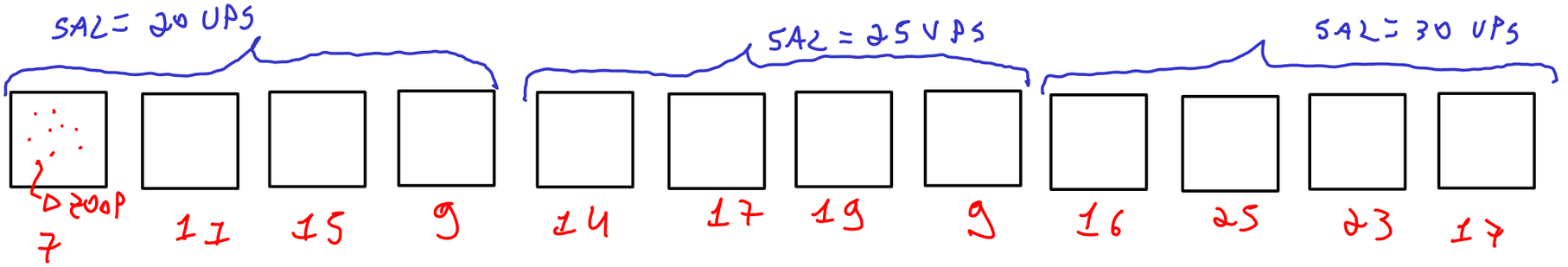
DIFERENTES ABORDAGENS P/ ESTUDO DE GRADIENTE

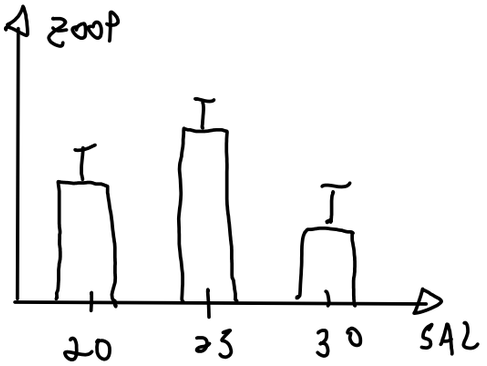
→ ABORDAGEM EXPERIMENTAL (ANOVA) → GRADIENTE CATEGORIZADO

EX: EFEITO DA SALINIDADE NO ZOOPLÂNCTON

↳ VAR. INDEPENDENTE ↳ VAR. DEPENDENTE



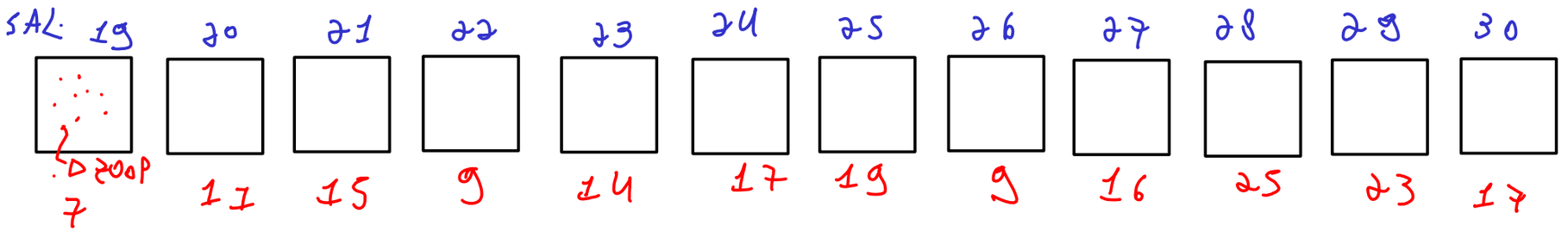
REP.	TRAT. 20	25	30
1	7	14	16
2	11	17	25
3	15	19	23
4	9	9	17



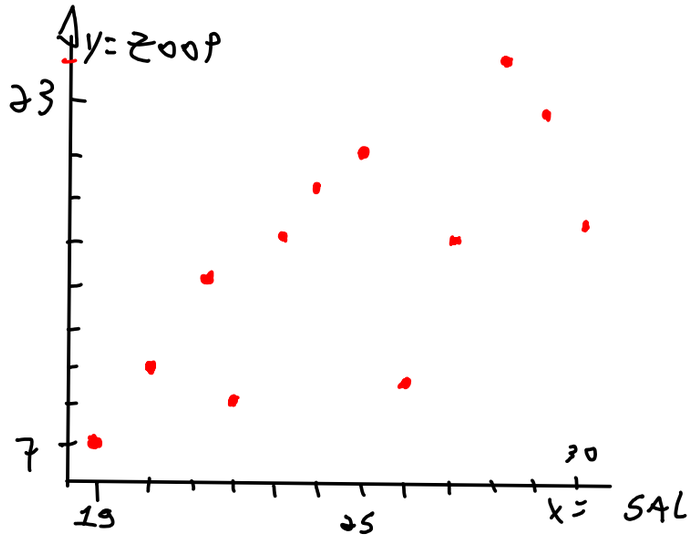
ANOVA

$$\begin{cases} H_0: \mu_1 = \mu_2 = \mu_3 \\ H_1: \text{pelo menos 1 é } \neq \end{cases}$$

→ ABORDAGEM CORRELATIVA (CORRELAÇÃO) → GRADIENTE CONTÍNUO



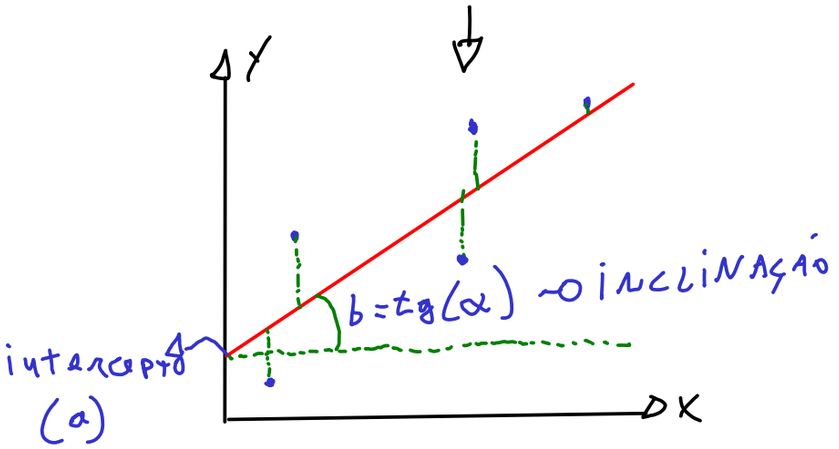
CRIAR UM DIAGRAMA DE DISPERSÃO



$$ZOOP = a + b \times SAL + E$$

REGRESSÃO E CORRELAÇÃO

MELHOR AJUSTE LINEAR: MÉTODO DOS MÍNIMOS QUADRADOS (LEAST SQUARE)

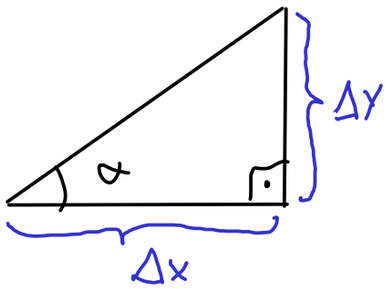


EQUAÇÃO DE REGRESSÃO

$$Y = a + b.X + E$$

E = ERRO OU RESÍDUO

CALCULANDO a e b



$$tg(\alpha) = b$$

$$b = \frac{\Delta Y}{\Delta x}$$

$$y = a + bx$$

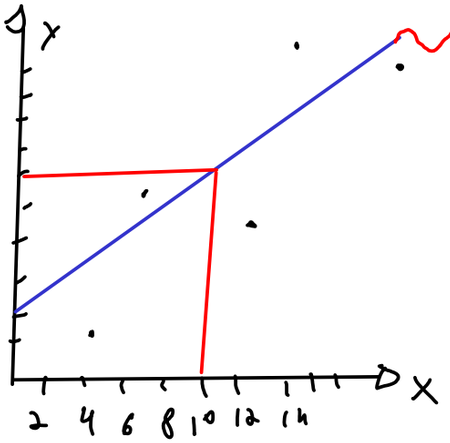
CALCULANDO a

$$\bar{y} = a + b\bar{x}$$

$$a = \bar{y} - b\bar{x}$$

VANTAGENS DA ABORDAGEM CORRELATIVA:

- USO DO GRADIENTE CONTÍNUO
- POSSIBILIDADE DE FAZER PREDIÇÕES



$y = a + bx$

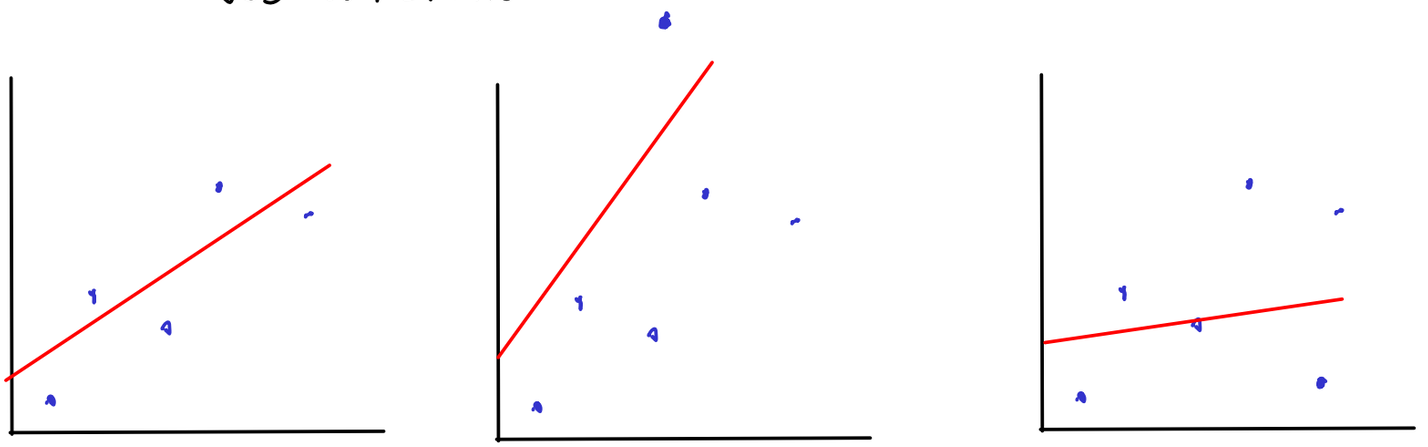
$a = 0,5 \quad b = 2$

PREDIZENDO O VALOR DE Y A PARTIR DE UM DADO VALOR DE X

SE $x = 10$ ENTÃO

$y = 0,5 + 2 \times 10$

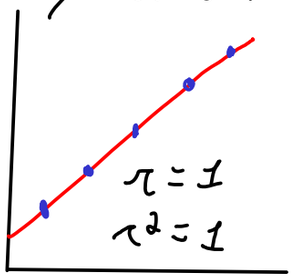
EFEITO DOS OUTLIERS



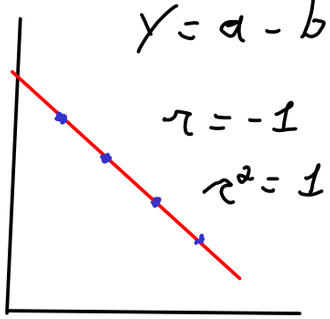
CORRELAÇÃO : GRAU DE AJUSTE

↳ r (PEARSON) $-1 < r < 1$

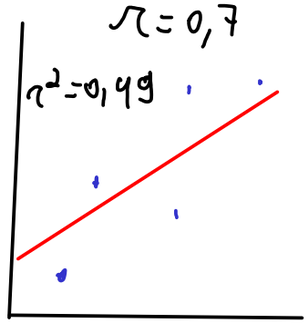
$y = a + bx$



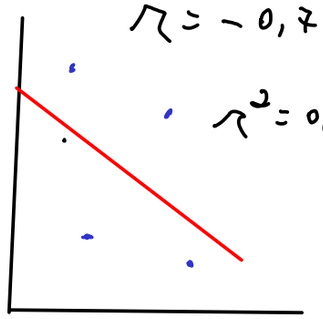
$y = a - bx$

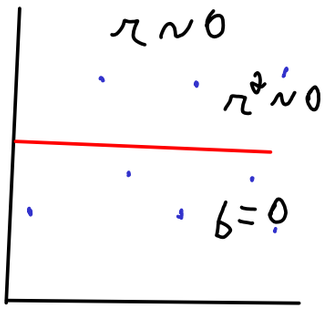


$r = 0{,}7$



$r = -0{,}7$





r = coef. corr $-1 < r < 1$

r^2 = coef. determinação $0 < r^2 < 1$

↳ um valor percentual.

INFERÊNCIA NA REGRESSÃO

Lembrando teste-t

$$t = \frac{\bar{x}_1 - \bar{x}_2}{EP_{12}}$$

NA REGRESSÃO:

$$t = \frac{b - 0}{EP_b}$$

$$\begin{cases} H_0: b = 0 \\ H_1: b \neq 0 \end{cases}$$

OUTPUTS DAS ANÁLISES

a b r r^2 p n

PRESSUPOSTO DA ANÁLISE:

NORMALIDADE DOS RESÍDUOS